

Rates of Change and Rectilinear Motion

ANSWER KEY



Position, velocity, and acceleration

- 1 A particle moves along a line with position $s(t) = t^3 - 6t^2 + 9t$ (metres, seconds).
 - a Find the velocity function $v(t)$. $v(t) = 3t^2 - 12t + 9$.
 - b Find the acceleration function $a(t)$. $a(t) = 6t - 12$.
 - c Find $v(2)$ and $a(2)$. $v(2) = 12 - 24 + 9 = -3$ m/s; $a(2) = 12 - 12 = 0$ m/s².
 - 2 For the particle above, find all times $t \geq 0$ when the particle is at rest ($v(t) = 0$).
 $3t^2 - 12t + 9 = 0 \Rightarrow t^2 - 4t + 3 = 0 \Rightarrow (t - 1)(t - 3) = 0$: $t = 1$ and $t = 3$.
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Speeding up and slowing down

- 3 Using the particle from above, determine whether the particle is speeding up or slowing down at $t = 2$ (compare the signs of $v(2)$ and $a(2)$).
 $v(2) = -3$ (negative) and $a(2) = 0$. Since $a(2) = 0$, the particle is neither speeding up nor slowing down at that instant — it is at a local extremum of speed.
 - 4 Determine whether the particle is speeding up or slowing down at $t = 0$.
 $v(0) = 9 > 0$; $a(0) = -12 < 0$. Since v and a have opposite signs, the particle is slowing down at $t = 0$.
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Total distance vs. displacement

- 5 For the particle above, find its displacement over $[0, 3]$.
Displacement = $s(3) - s(0) = (27 - 54 + 27) - 0 = 0$ m.
 - 6 Using the fact that the particle changes direction at $t = 1$ (found earlier), find the total distance traveled over $[0, 3]$.
 $s(0) = 0$, $s(1) = 1 - 6 + 9 = 4$, $s(3) = 0$. Distance
 $= |s(1) - s(0)| + |s(3) - s(1)| = |4 - 0| + |0 - 4| = 4 + 4 = 8$ m.
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Related rates

- 7 A spherical balloon is inflated so its radius grows at 2 cm/s. Find the rate of change of its volume when $r = 6$ cm, using $V = \frac{4}{3}\pi r^3$.
 $\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt} = 4\pi(36)(2) = 288\pi \approx 904.8$ cm³/s.

- 8 A 13 m ladder leans against a wall. The base slides away from the wall at 3 m/s. Find how fast the top of the ladder is sliding down when the base is 5 m from the wall.
- $x^2 + y^2 = 169$: $2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0$. At $x = 5$: $y = 12$. $5(3) + 12 \frac{dy}{dt} = 0$, so $\frac{dy}{dt} = -\frac{15}{12} = -1.25$ m/s (sliding down at 1.25 m/s).

Motion from a velocity graph

- 9 A particle's velocity is $v(t) = t^2 - 4t$ (m/s) for $t \geq 0$. Find the times when the particle changes direction.
- $v(t) = t(t - 4) = 0$: $t = 0$ and $t = 4$. The particle changes direction at $t = 4$ (velocity changes sign from negative to positive); $t = 0$ is just the start.
- 10 For the same particle, find the acceleration at $t = 1$ and state whether the particle is speeding up or slowing down at that instant.
- $a(t) = 2t - 4$; $a(1) = -2$. $v(1) = 1 - 4 = -3$ (negative). Since $v(1)$ and $a(1)$ have the same sign (both negative), the particle is speeding up at $t = 1$.

Visualizing position, velocity, and acceleration together

- 11 The graph shows the position function $s(t) = t^3 - 6t^2 + 9t$ from the first problem. Use the graph to identify the interval(s) where the particle is moving to the left (i.e. s is decreasing).
- The graph decreases on $(1, 3)$ (between the two points found earlier where $v(t) = 0$), so the particle moves to the left (negative direction) during that interval.

An applied related-rates capstone

- 12 A plane flies at a constant altitude of 3 miles, moving horizontally at 500 mph, directly away from a radar station on the ground. Find the rate at which the distance between the plane and the radar station is increasing when the plane is 4 miles horizontally from the station (directly above a point 4 miles from the station).
- Let x = horizontal distance, s = distance from radar (with height fixed at 3): $s^2 = x^2 + 9$. $2s \frac{ds}{dt} = 2x \frac{dx}{dt}$. At $x = 4$: $s = \sqrt{16 + 9} = 5$. $5 \frac{ds}{dt} = 4(500) = 2000$, so $\frac{ds}{dt} = 400$ mph.

Motion with a velocity given as a formula

- 13 A particle's velocity is $v(t) = 6t - t^2$ (m/s) for $0 \leq t \leq 8$. Find the particle's position function $s(t)$ given $s(0) = 0$, then find its displacement over $[0, 6]$.
- $s(t) = \int v(t) dt = 3t^2 - \frac{t^3}{3} + C$. Using $s(0) = 0$: $C = 0$. $s(t) = 3t^2 - \frac{t^3}{3}$. Displacement $= s(6) - s(0) = (108 - 72) - 0 = 36$ m.

A capstone rectilinear motion problem

- 14 A rocket's height is $h(t) = -5t^2 + 40t$ (meters, seconds) after launch. Find its maximum height, and the time at which it returns to the ground.

$h'(t) = -10t + 40 = 0$ gives $t = 4$ (time of maximum height). $h(4) = -80 + 160 = 80$ m. Ground return: $h(t) = 0$ gives $-5t(t - 8) = 0$: $t = 0$ or $t = 8$ s.